Module 2

Univariate
- Categorical
  - Bar charts
  - Pie charts
- Continuous
  - Histogram
  - Boxplot*
  - Density plot*
  - Violin plot**

大部分对应：measure → continuous, category → discrete

Standard visualisation for univariate

→ create a dimension (binned)

Bivariate
- Continuous → measure / measure
  - Scatter plot
  - Correlation matrix*
- Categorial → category / category
  - Contingency table
  - Density plot*
- Category/Numeric
  - Side-by-side boxplots
  - Side-by-side violin plots**

- { measure names (Dimension)：横位的名字.
  measure values (Measures)：横位下的各数值.
- Hierarchies.
- Date and Time { hierarchies.
  discrete / continuous? depends.
- Georgraphic

Tableau recommend use "relationship"

- Combining Data. → first, use the others if needed.
  - Union = 合并 rows (常见于合并不同日期，不同category的表)
  - Join = 合并 column
  - Relationships: flexible way of combine.

    优：Related Table remain seperate (not aggregated to single Table)

Logical Layer	Physical Layer
Relationships canvas in the Data Source page	Join/Union canvas in the Data Source page
Tables that you drag here are called logical tables	Tables that you drag here are called physical tables
Logical tables can be related to other logical tables	Physical tables can be joined or unioned to other physical tables
Logical tables are like containers for physical tables	Double-click a logical table to see its physical tables
Level of detail is at the row level of the logical table	Level of detail is at the row level of merged physical tables
Logical tables remain distinct (normalized), not merged in the data source	Physical tables are merged into a single, flat table that defines the logical table

Module 3

Operator/Keyword	Description
AND, OR, NOT	Standard Boolean operators
IN	Test for equality of a value against list of values
= or ==	Logical test of equality (= and == are equilvalent)
+	Addition of numeric or date values or concatenation of strings
-	Subtraction of numeric or date values
*, /	Multiplication or division of numeric values
^	Raise to a power with numeric values
()	Define order of operations or enclose function arguments
[]	Enclose field names
{}	Enclose level of detail calculations
//	Start a comment

- Row level Calculation.

- Transform skewed data: log → ln([ ])
- Dates, times 本身是 dimension.

  time between dates 是 measure =

  DATEDIFF('時間格式', [ ], [ ])

- Aggregate Calculation.

Aggregation	Calculation Description
Sum	Shows the sum of the values shown in the row or column.
Average	Shows the average of the values shown in the row or column.
Median	Shows the median for the values shown in the row or column.
Count; Count Distinct	Shows how many values or distinct values are displayed in the rows and columns in the view.
Minimum	Shows the minimum value shown in the row or column.
Maximum	Shows the maximum value shown in the row or column.
Percentile	Shows the average percentile for all values shown in the row or column.
Standard Deviation	The grand total using standard deviation is the standard deviation of the values shown in the row or column.
Variance	The grand total using variance is not the variance of the rows and columns in which they reside but rather of the underlying data behind the row or column.

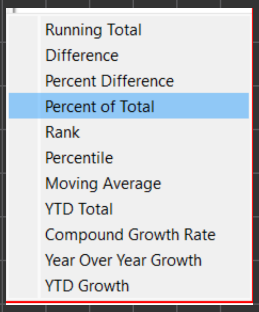
- Parameter - 使用者可控制的私密變數.
  - Reference 變數.
  - 可放 string.
  - 可當門檻. (Threshold)

{ Aggregate Calculation 對整個 data set 做計算. Result 作為 visualisation.
Table Calculation 對 sheet 上的 data 進行計算 ↓後 | 前

- Table Calculation 對現有 sheet 上的 Table 的
  - Partitioning = "Scope" of the calculation.
  - Addressing = "Direction" of the calculation.

specification level ⊖ ⊕

① Quick table Calculation ⟶



② Field editor, with standard table editor setting. (across, down)

③ Field editor, with specific dimensions.

↳ { ☑ addressing: 用以此我的單位區
☐ Partition: 區分重新計算的各區

②

- Table reference functions
  - INDEX()
  - FIRST()
  - LAST()
- Access other values in table
  - LOOKUP(expression, [offset])
  - PREVIOUS_VALUE(expression)
  - NEXT(expression)
- Return aggregates of the whole partition
  - SIZE()
  - TOTAL()
  - etc.

只有這兩者是 Non-direction Function.

- "Rank" functions
  - RANK(expression, ['asc' | 'desc'])
  - RANK_DENSE
  - RANK_UNIQUE
  - etc.
- "Running" functions
  - RUNNING_AVG(expression)
  - RUNNING_COUNT(expression)
  - RUNNING_MIN(expression)
  - RUNNING_SUM(expression)
  - etc.
- "Window" functions
  - WINDOW_AVG
  - WINDOW_SUM
  - WINDOW_MAX
  - etc.

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① Index ( ) 值的位置 (partition 內)

② FIRST ( ) 值 ⟷ 第一個值加總的數量

③ LAST ( ) 值 ⟷ 最後一個值 〃

④ SIZE ( ) 欄位裡的總數

⑤ TOTAL ( ) 總值.

往回 ⊖
往前 ⊕

⑥ LOOKUP (expression, offset)
table cal / aggre cal

⑦ PREVIOUS_VALUE() 上一個值

⑧ RANK
超常用!
⑨ RANK_DENSE(expression, ['asc'/'desc'])
⑩ RANK_MODIFIED "
⑪ RANK_PERCENTILE "
⑫ RANK_UNIQUE "

有tie跳号.
不跳号.
名次前面有n人
百分比.
每人名次不重复

⑬ RUNNING_AVG(expression)
⑭ RUNNING_COUNT
⑮ RUNNING_MAX
⑯ RUNNING_MIN
⑰ RUNNING_SUM

到目前為止的這一行.
(依賴視圖順序)

⑱ WINDOW_AVG(expression, [start, end])
⑲ _CORR "
⑳ _COUNT "
㉑ _MAX "
㉒ _MIN "
㉓ _MEDIAN "
㉔ _PERCENTILE "
㉕ _STDEV "
㉖ _STDEVP "
㉗ _SUM "
㉘ _VAR "
㉙ _VARP "
㉚ _COVAR(exp1, exp2, [start, end])
㉛ _COVARP "

㉜ DATEPART('datepart', [ ]) 取某部份.

㉝ DIFF('', [start], [end])

㉞ ADD('', number, [ ]) 增加.

㉟ TRUNK('', [ ]) 截斷.

* compound growth rate.

$$CAGR = \left(\frac{\text{ending value}}{\text{Beginning value}}\right)^{\frac{1}{n}} - 1$$

$$= \frac{ZN(SUM([sales]))}{LOOKUP(ZN(SUM([sales])), first())} - 1$$

③ specific dimension

- ☑ checked = defines the addressing (計算的方向)
- ☐ unchecked = defines the partitioning.

— Level of Detail Calculation. (LOD)

只用指定的層級來計算. 常用於以某些基準計算

① FIXED [Dimension] : AGG([Measure])

只看指定的維度.

② INCLUDE [Extra Dimension] = AGG([Measure])

新增維度後再算

③ EXCLUDE [Dimension] = AGG([Measure])

忽略某些層級再算

Mid - Prep

Q8 Change from previous month

ZN(SUM([ ]) - LOOKUP(ZN(SUM([ ]), -1)

Q10 Time to ship.

DATEDIFF('日期粒度', [start], [end])

Q12 if ____ OR ____ THEN ____ ELSE ____ END.

從開頭一路累加到現在.

Q13 Running Sum

RUNNING_SUM(SUM([sales]))

最近n筆的.

Q16 Moving Average.

numbers.

WINDOW_AVG(SUM([ ]), start, end)

Q18 Rank

|

Q20 RANK([ ])

Module 4

USC Viterbi School of Engineering

PRIMARY DASHBOARD TYPES
COMPARISON MATRIX

Dashboard Type	Primary Purpose	Typical Users	Decision Timeframe	Update Frequency	Interactivity Level
Operational	Monitor ongoing operations and respond immediately	Front-line staff, operations managers	Minutes to hours	Real-time or near-real-time	Low–Moderate
Tactical	Manage short- to mid-term execution and adjustments	Managers, team leads	Days to weeks	Daily or weekly	Moderate
Strategic	Track long-term performance against goals	Executives, senior leadership	Months to years	Weekly, monthly, quarterly	Low
Analytical	Explore data to identify patterns and causes	Analysts, data scientists	Variable / exploratory	On demand or periodic	High
Informational	Communicate key facts and summaries	Broad or external audiences	Contextual / reference	Periodic	Very low

Module 6

教她何用 console / python-code 建立 model / prediction (mostly ad-hoc)

Console-Based ML

What AutoML (console) gives us:

- Fast baseline models
- Built-in evaluation
- Deployment in minutes
- Integrated explanations

What we don't control (yet)

- Feature engineering logic
- Preprocessing choices
- Experiment structure
- Reproducibility

Python Notebook Based AutoML

Client library - primary component.

- Datasets and data labeling
- AutoML
- Custom training and models
- Model evaluation and experiment tracking (including explainable AI)
- Hyperparameter tuning
- Model deployment and prediction (online, batch, and edge)
- MLOps tools
  - Pipelines
  - Feature store
  - Model registry
  - Model monitoring

start P. 156.

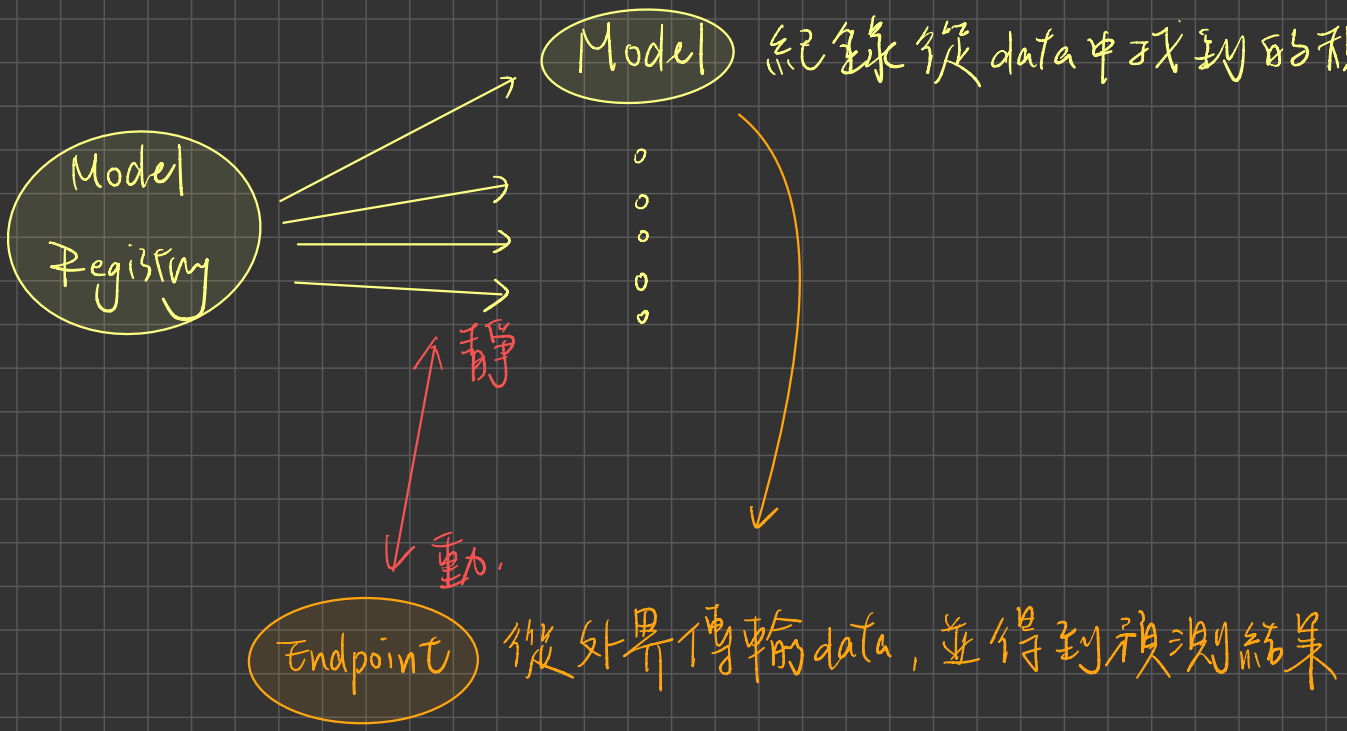
① Setup (Import library, configure parameter, Initialize Clients)
② Create a Vertex AI dataset (link to Big Query)
③ Retreive model
④ Evaluation → 使用前確認模型測能力如何.
⑤ create endpoint
⑥ deploy endpoint
⑦ Train AutoML
⑧ evaluate model

Customed Model

start P. 213.

① Setup (Import library, configure parameter, Initialize Clients)
② Create a Vertex AI dataset (link to Big Query)
③ Retreive model
④ Evaluation
⑤ create endpoint.
⑥ Train AutoML → Data Prep.
{ de skew
Split data.
⑦ Deploy model to endpoint → { preprocessing. P. 228
pickle file → pipeline.
upload model to registry.
deploy to endpoint
⑧ evaluate model

觀念釐清


Model
紀錄從data中找到的規律.
Model
Registry
靜
動.
Endpoint
從外界傳輸data，並得到預測結果

Module 7 介紹MLOps = 系統化ML, 及相關功能.

簡介MLOps

(data變 → model壞)

- 痛点 = model最難的是後續的維運
- 解方 = MLOps　　重複 operation / process.

MLOPS

New Capabilities.

① Experiment Tracking.
　紀錄. → 成為可管理的實驗系統.

② Pipeline Automation
　→ 可重複執行的

③ Model Registry.
　解決 Train-Serving skew 的問題.

④ Monitoring.
　Drift detection / scheduled retrain. → data drift 系統自己變差

核心功能.

① Automation 自動化.

② Collaboration 協作

③ Reproducibility. 可重現

- ① Functional reproducibility. = input相同 → output相同.
  Data Split / Preprocessing / Model 等都要相同

三層次

② Data reproducibility. = 資料本身可控.

③ Operational reproducibility. = retain (保存)

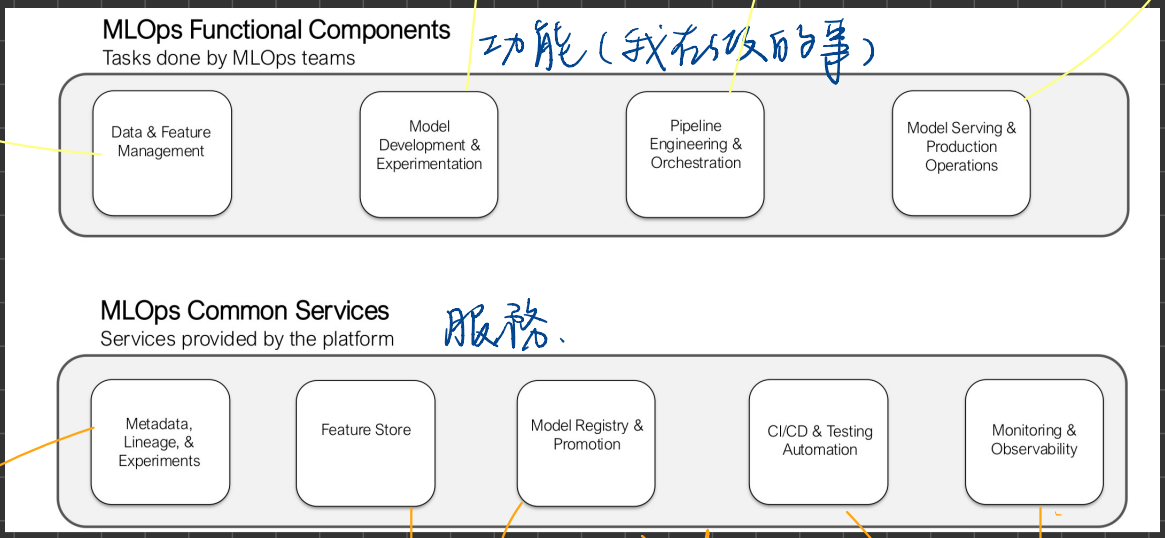
Architecture

- collect data
- data cleaning.
- feature engineering.

- 選 feature
- hyperparameter tuning.
- 比較

- deploy model
- version
- Scaling.

- pipeline
- manage workflow



- 紀錄 feature / model
- 存 hyperparameter

versioned lifecycle (dev → Test → prod)

依賴 feature

- 自動化 (pipeline)
- validation

- Monitor Model / Data / System

Module 8 model development/experimentation.

layers of decisions. (多維度)

- Model decisions (logistic regression? random forest? XGBoost?)
- Hyperparameters (learning rate, tree depth, number of trees, etc.)
- Feature decisions (which predictors? engineered features?) → tt model 重要
- Data partitioning (train/test split, cross-validation, sampling strategy

experiment 要紀錄的事情.

Model configuration
- Model type / algorithm
- Hyperparameter values
- Training configuration

Feature set
- Variables used in the model
- Feature engineering steps
- Feature dataset or feature view

Data partitioning
- Training / validation / test split
- Sampling strategy
- Dataset version

Performance metrics on validation/test data

Artifacts
- Trained model
- Prediction outputs
- Plots or evaluation reports

Module 9 Pipeline Engineering & Orchestration.

Pipeline

- data → feature → train → evaluate
- 四個目標
  - Automation
  - Consistency.
  - Scalability.
  - Reliability.

T	P
0	1
0	0
0	1
0	0
1	1
0	0
1	1
0	0
0	1
1	0

- ① Training Pipeline

> - Purpose: create and update models
> - Typical steps
>   - Data ingestion
>   - Data preparation / feature engineering
>   - Train/validation split
>   - Model training
>   - Evaluation
>   - Model registration
> - When it runs
>   - On schedule
>   - Triggered by new data
>   - Triggered by experimentation

② Inference Pipeline

> - Purpose: generate predictions from a trained model
> - Typical steps:
>   - Input data ingestion (real-time or batch)
>   - Feature transformation (must match training!)
>   - Model prediction
>   - Post-processing (e.g., thresholds, formatting)
>   - Output delivery (API, dashboard, database)

Data Leakage

- cause
  - ① Response variable is a function of predictors.
    ex: GDP per capital v.s. GDP. 有關.
  - ② Feature from the future
    ex: 是否選秋 v.s. 是否會選我.

- Type
  - Partition Leakage
    資料同時在 train / test.
  - Imputation Leakage
    不能用 training set 做 Imputation.

Imbalance data

Sampling.

Feature Engineering.

- Transform raw data → Prediction vector.
- Good Feature
  1. Predictive Power
  2. Fast Computability. (計算速度快)
  3. Uncorrelatedness 不相關性 (可靠性)
- data type
  - text
    - one hot encoding: 類別轉為一進位,
    - bag of words: 文字轉為一進位
  - binary.
    - Discretization (分類)
      - uniform
      - k-means.
      - quantile.
- Best Practice.

› Generate many simple features
› Reuse legacy models (use outputs as features)
› Use IDs as features if they represent natural groupings of data (e.g., zip code or city ID)
› Don't use unique IDs as features
› Reduce cardinality when possible
  » Group similar features
  » Group the "long tail"
  » Don't use features where all values are unique or one value dominates
› Use counts with caution (e.g., "number of calls since subscription")
  » Potential for data leakage, risk of high cardinality, scaling and interpretation vchallenges
› Keep code, model, and data in sync

Model Deployment

① static deployment
模型封装在应用程式中，
跑的速度快，但更新麻烦。

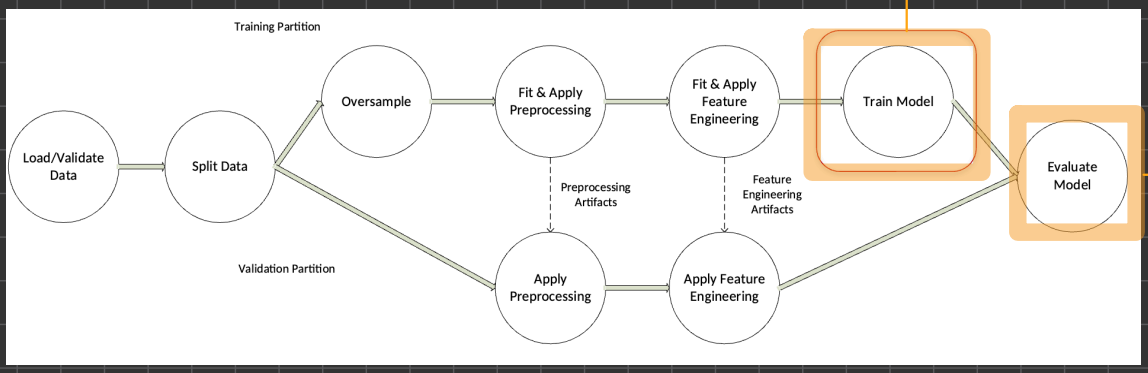
② Dynamic Deployment on user
模型和代码分离，可单独更新参数。

common
③ 〃 on Server.
模型运行在服务器上。

Training Pipeline

輸出：Artifact, Model

多3這麼個步驟.

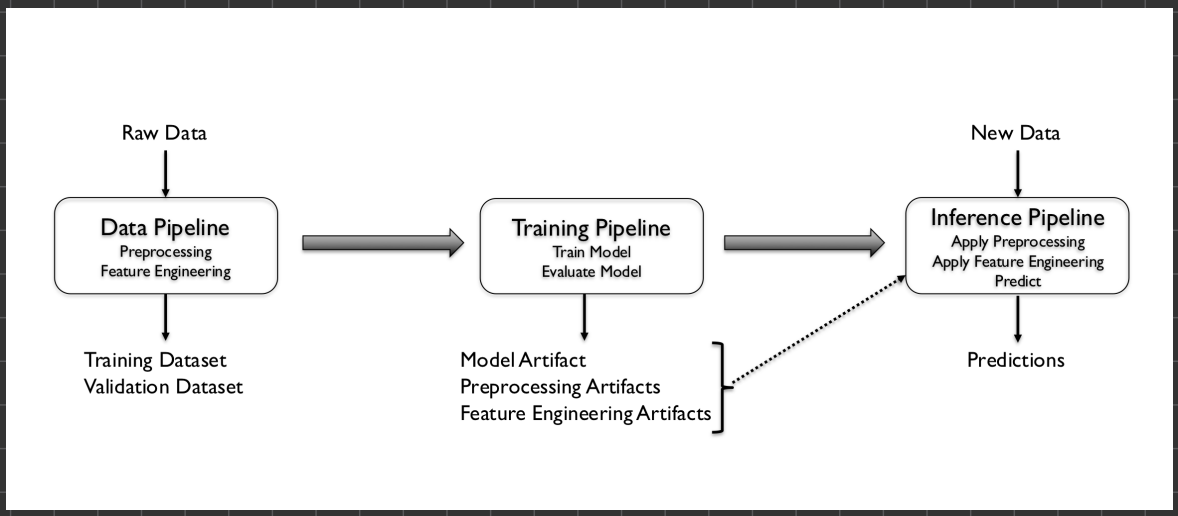


Inference Pipeline

使用已經訓練好的model做預測.

- No learning occurs during inference
- All transformations are:
  - precomputed
  - reused from artifacts
- Model + preprocessing + feature engineering form a single unit

Inference is replay, not learning


Raw Data
Data Pipeline
Preprocessing
Feature Engineering
Training Dataset
Validation Dataset
Training Pipeline
Train Model
Evaluate Model
Model Artifact
Preprocessing Artifacts
Feature Engineering Artifacts
New Data
Inference Pipeline
Apply Preprocessing
Apply Feature Engineering
Predict
Predictions